Points: 13.5/16 (84.38 %)

Algebraic numbertheory
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Seite 2.

Aufgabe 1) Let $\varphi = \frac{\sqrt{5}+1}{2} \in \mathbb{C}$

(i) t.s. φ is integral over $\mathbb{Z}$.

proof It holds that $\varphi^2 = \frac{5+1+2\sqrt{5}}{4}$, thus

$\varphi^2 - \varphi - 1 = 0$, hence φ is the root of

$P = T^2 - T - 1 \in \mathbb{Z}[T]$ which is monic, thus we can conclude. ☑ ok

(ii)

t.s. φ^{-1} is integral over $\mathbb{Z}$.

proof We know that $\varphi^2 - \varphi - 1 = 0$, thus.

$$\varphi(\varphi - 1) = 1.$$

And since $\mathbb{C}$ is a commutative ring, we get that $\varphi^{-1} = \varphi - 1$. Furthermore is $\varphi^{-1} = \varphi - 1$ integral over $\mathbb{Z}$, since the integral elements form a ring. (and φ, 1 are) ☑ ok

↳ since a root of $X - 1$.

(iii)

t.s. $\mathbb{Z}[\varphi]$ is not finite.

proof. We denote $M = \{\varphi^n \mid n \in \mathbb{N}\} \subseteq \mathbb{Z}[\varphi]$.

claim 1 $M \subseteq \mathbb{Z}[\varphi]^\times$

claim 2 M is not finite.

By claim 2 M is not finite and by $M \subseteq \mathbb{Z}[\varphi]^\times$ we get that $\mathbb{Z}[\varphi]^\times$ is neither finite and we can conclude. ✓

It is left to prove the claims.

Let $n \in \mathbb{N}$, then it holds that

$$\varphi^n \cdot (\varphi - 1)^n = 1, \text{ where } (\varphi-1)^n \in \mathbb{Z}[\varphi] \text{ since } \varphi - 1 \in \mathbb{Z}[\varphi]$$

Thus $\varphi^n \in \mathbb{Z}[\varphi]^\times$ and the claim 1 follows. ✓

We assume that M is finite. But then, there exists $m, n \in \mathbb{N}$, s.t. $\varphi^m = \varphi^n$. w.l.o.g. we may assume $m > n$

This implies that

$$\varphi^{m-n} = 1. \quad \text{For } q = m - n > 0.$$

But for any $l \in \mathbb{N}$ it holds that

$$\varphi^l = \frac{1}{2}\sum_{k=0}^{l} \binom{l}{k} (\sqrt{5})^k > 1 \quad (\text{since } l \geq 2,\ k \geq 2)$$

→ why?

Thus this is by assumption a contradiction, thus there can not exist such m, n hence M is not finite. ☑

1.75

Ex 2.

Let $\varphi = \frac{3+4\sqrt{-1}}{5}$

t.s. φ is not integral over $\mathbb{Z}$.

proof We assume that φ is integral over $\mathbb{Z}$. ok

But then, there exists a $n \in \mathbb{N}$ s.t.

$$\mathbb{Z}[\varphi] = \varphi\mathbb{Z} + ____ + \varphi^n\mathbb{Z}$$ ok

Thus there exist $a_0, ___, a_n$, s.t.

$$\frac{(3+4\sqrt{-1})^{n+1}}{5^{n+1}} = \varphi^{n+1} = a_0 + a_1\varphi + ____ + a_n\varphi^n$$

By multiplying with 5^n, we get

$$5^n\varphi^n = \frac{(3+4\sqrt{-1})^{n+1}}{5} = \frac{3^{n+1}}{5} + \sum_{k=1}^{n} \frac{3^{n-k}(4\sqrt{-1})^k}{5}\binom{n}{k} = 5^n a_0 + 5^{n-1}(3+4\sqrt{-1})a_1 + ____ + (3+4\sqrt{-1})^n a_n$$

false: $\sqrt{-1} \in \mathbb{Z}$

$n=1$: $\frac{(3+4i)^2}{5} = \frac{9-16+24i}{5} = \frac{9}{5} + \frac{-16+24i}{5}$, $\frac{24}{5} \notin i\mathbb{Z}$

But then, it holds that $\longrightarrow$ $5^2\varphi^2 = \frac{9}{5} + \frac{-16+24i}{5} = \underbrace{-1}_{\in\mathbb{Z}} + \frac{24i}{5}$

$$\mathrm{Re}(5^n\varphi^n) \overset{\text{false}}{=} \frac{3^{n+1}}{5} \in \mathbb{Q}\setminus\mathbb{Z}$$ since 3, 5 coprime

and $\mathrm{Re}(5^n a_0 + 5^{n-1}(3+4\sqrt{-1})a_1 + ____ + (3+4\sqrt{-1})^n a_n)$

$$= \sum_{k=0}^{n} 5^{n-k}\cdot 3^k a_k \in \mathbb{Z}$$

which is absurd. Hence φ can not be integral. ☐

The argument doesn't work

0.5/2

Ex 3.)

Let K/F be a finite field extension.

(i) t.s. $\mathrm{Tr}_{K/F}(\lambda x + \mu y) = \lambda \mathrm{Tr}_{K/F}(x) + \mu \mathrm{Tr}_{K/F}(y)$ for any $x, y \in K$, $\lambda, \mu \in F$.

proof. Let $x, y \in K$, $\lambda, \mu \in F$.

Since K/F is finite, there exists a basis $\mathcal{B}$ of K as a F-vectorspace. We denote $n = \dim_F(K)$.

Then, from linear algebra, we know that

$K \simeq K^* \simeq \mathrm{Mat}_n(F)$ are canonically isomorph. as F-vectorspaces.

i.e. we get the ~~a~~ isomorphism

$$K \longrightarrow \mathrm{Mat}_n(F), \quad x \longmapsto \varphi_x.$$

Since this is in particular linear, we get by $\lambda \mathrm{Tr}(A) = \mathrm{Tr}(\lambda A)$, $\mathrm{Tr}(A) + \mathrm{Tr}(B) = \mathrm{Tr}(A+B)$ for any $A, B \in \mathrm{Mat}_n(F)$ that

$$\lambda \mathrm{Tr}_{K/F}(x) + \mu \mathrm{Tr}_{K/F}(y) \overset{\checkmark}{=} \lambda \mathrm{Tr}(\mathrm{Mat}_{\varphi_x, \mathcal{B}}) + \mu \mathrm{Tr}(\mathrm{Mat}_{\varphi_y, \mathcal{B}})$$

$$\overset{\checkmark}{=} \mathrm{Tr}(\mathrm{Mat}_{\varphi_{\lambda x + \mu y}, \mathcal{B}})$$

$$\overset{\checkmark}{=} \mathrm{Tr}_{K/F}(\lambda x + \mu y).$$

□ 2

(ii) t.s. $N_{K/F}(xy) = N_{K/F}(x) N_{K/F}(y)$

proof. By linear algebra, we know that the determinant of a matrix is multiplicative. Hence, by the same arguments, we get

$$N_{K/F}(x) N_{K/F}(y) \overset{\checkmark}{=} \det(\mathrm{Mat}_{\varphi_x, \mathcal{B}}) \cdot \det(\mathrm{Mat}_{\varphi_y, \mathcal{B}})$$

$$\overset{\checkmark}{=} \det(\mathrm{Mat}_{\varphi_x, \mathcal{B}} \cdot \mathrm{Mat}_{\varphi_y, \mathcal{B}})$$

$$\overset{\checkmark}{\underset{(i)}{=}} \det(\mathrm{Mat}_{\varphi_{x \cdot y}, \mathcal{B}}) \overset{\checkmark}{=} N_{K/F}(xy).$$

□ 2

N B. For this ~~&~~ step, we used that the map $K \longrightarrow \mathrm{Mat}_n(F)$, $x \longmapsto \varphi_x$ is even ~~&~~ iso as a F-Algebra. This holds since a small computation shows that for any $x, y, z \in K$ we have

$$xy \cdot z = \varphi_x(z) \cdot \varphi_x(y) = \mathrm{Mat}_{\varphi_x, \mathcal{B}} \circ \mathrm{Mat}_{\varphi_y, \mathcal{B}} \cdot z.$$

Ex 4.)

Let $p>2$ be a prime number and let $\zeta = \exp\left(\frac{2\pi i}{p}\right)$

(i) t.s. $(1, \zeta, \dots, \zeta^{p-2})$ is a basis of $\mathbb{Q}(\zeta)$ as $\mathbb{Q}$-vectorspace.

proof. We know that $P = T^{p-1} + \dots + T + 1 \in \mathbb{Q}[T]$ is the minimal polynomial of ζ.

Thus we get

$$\zeta^p + \dots + 1 = 0 \text{ thus } \zeta^p = -1 - \zeta - \dots - \zeta^{p-1} \quad (1)$$

Hence $\zeta^p \in V = \operatorname{span}_{\mathbb{Q}}(1, \dots, \zeta^{p-1})$ and by multiplying (1) with ζ we get

$$\zeta^{p+1} = -\zeta - \zeta^2 - \dots - \underbrace{\zeta^p}_{\in V} \in V$$

and by continuing this inductively, we get $\zeta^n \in V \ \forall n \in \mathbb{N}$ and therefore $V = \mathbb{Q}[\zeta]$. Since ζ is algebraic over $\mathbb{Q}$, it holds that $\mathbb{Q}[\zeta] = \mathbb{Q}(\zeta)$ hence, we get $V = \mathbb{Q}(\zeta)$.

It is left to show that $(1, \zeta, \dots, \zeta^{p-2})$ is linearly indep.

Assuming, $(1, \zeta, \dots, \zeta^{p-2})$ is lin. dependent, there exist $b_0, \dots, b_{p-2} \in \mathbb{Q}$ not all 0, s.t.

$$b_0 + b_1\zeta + \dots + b_{p-2}\zeta^{p-2} = 0$$

Thus $f = \sum_{k=1}^{p-2} b_k T^k \in \mathbb{Q}[T] \setminus [0]$ with $\deg(f) < p$ and $f(\zeta) = 0$.

But we know $(P) = \ker(\mathbb{Q}[T] \to \mathbb{Q}[\zeta], T \to \zeta)$

Thus $f \in (P)$ which implies $P \mid f$ in $\mathbb{Q}[T]$ hence $\deg(P) < \deg(f)$

But since $\deg(P) = p-1$, this is absurd, so the lin. independency follows.

1.75

Ex 4.)

ii)

We denote $\varphi: \mathbb{Q}(\zeta) \longrightarrow \mathbb{Q}(\zeta), x \longmapsto \zeta x$. and

$B = (1, \dots, T^{p-2})$ the a basis of $\mathbb{Q}(\zeta)$.

Then it holds that

$$[\varphi(1)]_B = [\zeta]_B = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, [\varphi(\zeta^{p-2})]_B = [\zeta^{p-1}]_B = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

and $[\varphi(\zeta^{p-1})]_B = -1 - \dots - 1$ (By (1) of (ii).)

Thus we get

$$\mathrm{Mat}_{\varphi,B} = \begin{pmatrix} 0 & 0 & \cdots & 0 & -1 \\ 1 & 0 & & 0 & -1 \\ 0 & 1 & & 0 & -1 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & & 1 & -1 \end{pmatrix}$$ ✓

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iii) $\mathrm{Tr}_{\mathbb{Q}(\zeta)/\mathbb{Q}}(\zeta) = \mathrm{Tr}(\mathrm{Mat}_{\varphi,B}) = -1$ ✓

$N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(\zeta) = \det(\mathrm{Mat}_{\varphi,B}) = (-1)^{d} \cdot (-1) = ?$

changing the rows of that

what is d?

1.5